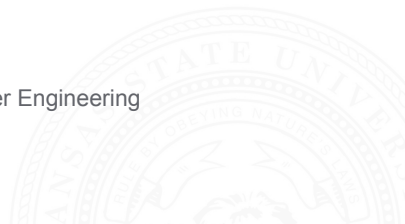


Per Unit System I

Normalization within one voltage level

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Lecture outline

- 1 Part 1: Why per unit?
- 2 Part 2: Bases and conversion
- 3 Part 3: Three-phase circuits
- 4 Part 4: Normalization and recovery
- 5 Part 5: Single-phase worked example

Specify bases, normalize the given data, solve in per unit, then recover physical quantities.

PART 1

Why per unit?

Relative scales, voltage levels, and equipment quantities

What is a per-unit system?

Express each electrical quantity as a ratio to a chosen base.

$$x_{\text{p.u.}} = \frac{x_{\text{actual}}}{x_{\text{base}}}$$

Actual voltage

218.5 kV

÷ 230 kV



Per-unit voltage

0.950 p.u.

0.950 p.u. means 95% of the chosen base.

The ratio is dimensionless. A value of 1.0 p.u. equals the chosen base.

Why use per unit?

Bus A

13.662 kV

Nominal: 13.8 kV



Bus B

505.0 kV

Nominal: 500 kV

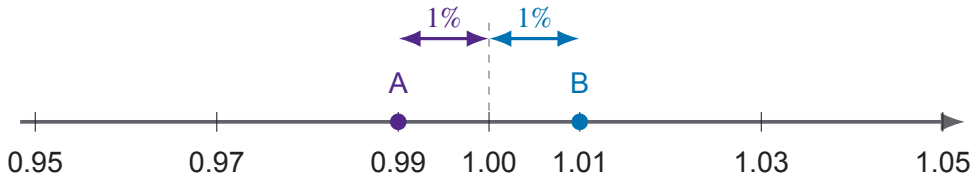


Which bus is closer to nominal?

Per unit makes voltage levels comparable

$$\text{Bus A: } 13.662/13.8 = 0.990$$

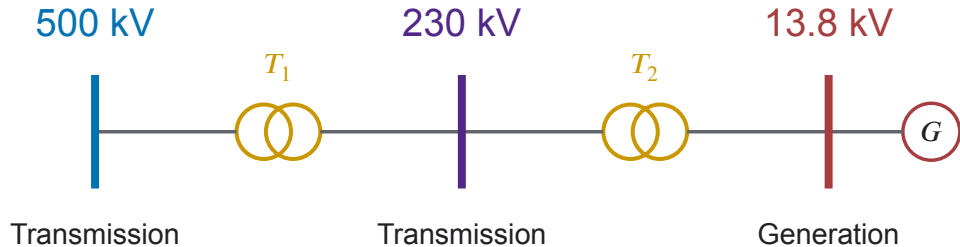
$$\text{Bus B: } 505/500 = 1.010$$



Nominal voltage is 1.00 on each local nominal base.

Both buses are equally close to nominal: one is below, the other above.

One network, several voltage levels



The same network contains several voltage levels.

A local voltage base lets us compare every bus on the same relative scale.

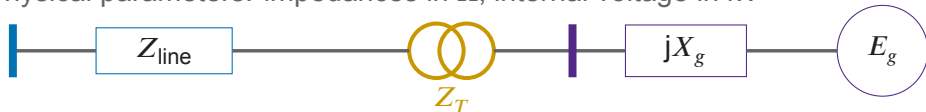
The same equipment in per unit

Line: 230-kV zone

Transformer

Generator: 13.8-kV zone

Physical parameters: impedances in Ω , internal voltage in kV



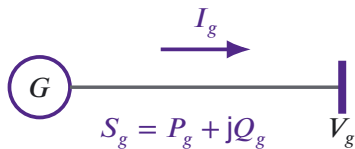
Per-unit parameters: the same equipment on relative scales



Generator, transformer, and line parameters all belong in the per-unit network model.

Equipment operating quantities in per unit

Generator terminals

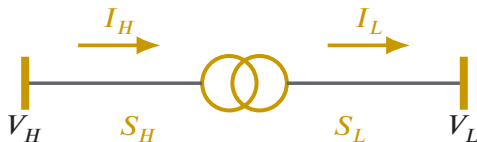


$$V_g \text{ (kV)} \mapsto V_{g,\text{p.u.}}$$

$$I_g \text{ (A)} \mapsto I_{g,\text{p.u.}}$$

$$S_g \text{ (MVA)} \mapsto S_{g,\text{p.u.}}$$

Transformer terminals



$$V_H, V_L \mapsto V_{H,\text{p.u.}}, V_{L,\text{p.u.}}$$

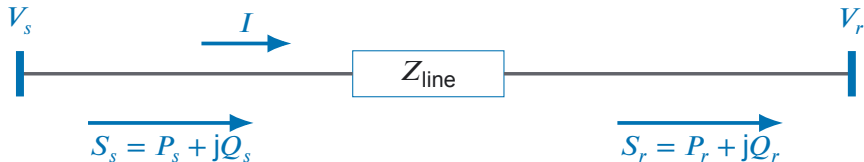
$$I_H, I_L \mapsto I_{H,\text{p.u.}}, I_{L,\text{p.u.}}$$

$$S_H, S_L \mapsto S_{H,\text{p.u.}}, S_{L,\text{p.u.}}$$

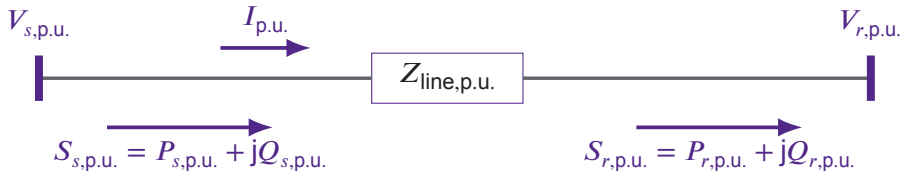
Operating voltage, current, and power have per-unit forms too, at every terminal.

Voltage, current, and power along a line

Physical quantities: kV, A, MVA, and Ω



Per-unit quantities: the same line and the same operating point



Next choose matching bases for V , I , S , and Z , then calculate their per-unit values.

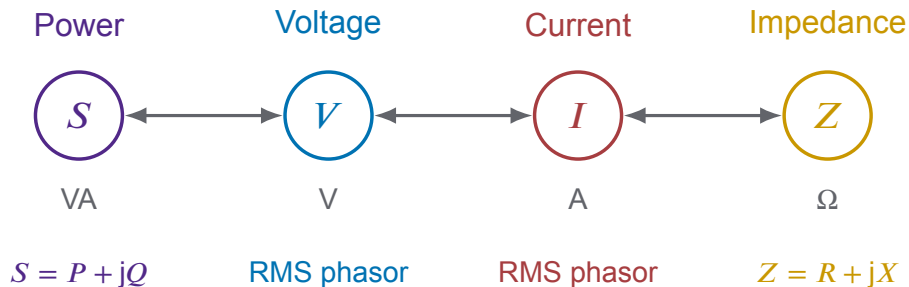
PART 2

Bases and conversion

Two independent bases, complex components, and conversion

Four coupled electrical quantities

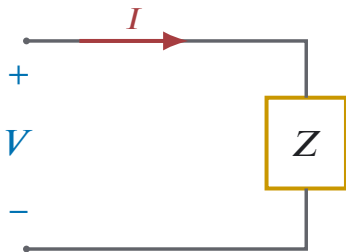
Four kinds of quantity describe the same electrical circuit.



The four quantities are coupled. Their bases must be coupled too.

Circuit laws couple the four quantities

Start with the voltage and current at the same circuit port.



Complex power

$$S = VI^*$$

Ohm's law

$$V = ZI$$

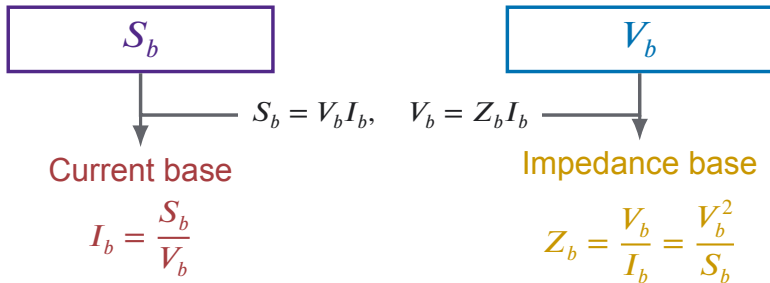
$S = P + jQ$ absorbed at the port

I^* : complex conjugate of I

The base scales must respect these same relationships.

Two independent bases determine the rest

Choose positive real power and voltage bases. Subscript b means base.



Four base values, only two independent choices.

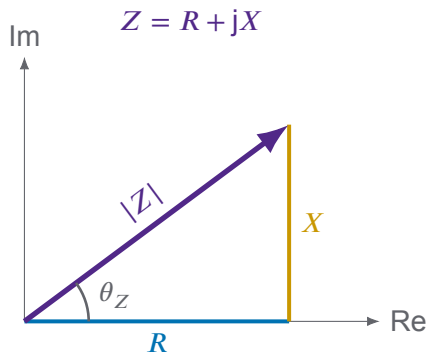
Choose S_b and V_b ; calculate I_b and Z_b . Do not choose all four independently.

Convert all four quantities

Physical quantity			Per-unit value
Power	S	$\div S_b \longrightarrow$	$S_{p.u.} = \frac{S}{S_b}$
Voltage	V	$\div V_b \longrightarrow$	$V_{p.u.} = \frac{V}{V_b}$
Current	I	$\div I_b \longrightarrow$	$I_{p.u.} = \frac{I}{I_b}$
Impedance	Z	$\div Z_b \longrightarrow$	$Z_{p.u.} = \frac{Z}{Z_b}$

$P_{p.u.} = P/S_b$ and $Q_{p.u.} = Q/S_b$: real and reactive power use the same power scale.

One impedance base for Z, R, and X



Use one positive real impedance base

$$\begin{aligned} Z_{\text{p.u.}} &= \frac{Z}{Z_b} = \frac{R + jX}{Z_b} \\ &= \frac{R}{Z_b} + j\frac{X}{Z_b} = R_{\text{p.u.}} + jX_{\text{p.u.}} \end{aligned}$$

$$R_{\text{p.u.}} = \frac{R}{Z_b}, \quad X_{\text{p.u.}} = \frac{X}{Z_b}$$

One real scale changes the size of the impedance triangle and preserves its angle. No separate R or X base is needed.

Admittance and its reciprocal base

Admittance in physical units

$$Y = \frac{1}{Z} = \frac{1}{R + jX}$$
$$= \frac{R - jX}{R^2 + X^2} = G + jB$$

$$G = \frac{R}{R^2 + X^2}, \quad B = \frac{-X}{R^2 + X^2}$$

Conductance G and susceptance B are in siemens, like Y .

One base for Y , G , and B

$$Y_b = \frac{1}{Z_b} = \frac{I_b}{V_b}$$

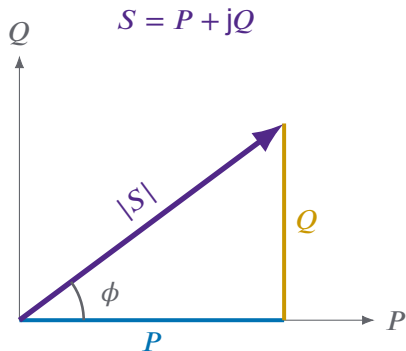
$$Y_{p.u.} = \frac{G + jB}{Y_b} = G_{p.u.} + jB_{p.u.}$$

$$G_{p.u.} = \frac{G}{Y_b}, \quad B_{p.u.} = \frac{B}{Y_b}$$

$$Y_{p.u.} = \frac{1/Z}{1/Z_b} = \frac{Z_b}{Z} = \boxed{\frac{1}{Z_{p.u.}}}$$

Y, G, and B share one admittance base. Take the reciprocal of the complete complex impedance.

One power base for P, Q, and S



Use one apparent-power base S_b

$$S_{p.u.} = \frac{S}{S_b} = \frac{P + jQ}{S_b}$$
$$= \frac{P}{S_b} + j\frac{Q}{S_b} = P_{p.u.} + jQ_{p.u.}$$

$$P_{p.u.} = \frac{P}{S_b}, \quad Q_{p.u.} = \frac{Q}{S_b}$$

Apparent power uses the same base:

$$|S_{p.u.}| = |S|/S_b.$$

P, Q, and S share the same numerical power base. The power triangle keeps its angle, so power factor is unchanged.

Coherent bases preserve circuit laws

Physical circuit

Per-unit circuit

$$S = VI^* \xrightarrow{\div S_b} S_{\text{p.u.}} = V_{\text{p.u.}} I_{\text{p.u.}}^*$$

$$\frac{VI^*}{S_b} = \frac{V}{V_b} \frac{I^*}{I_b} \quad \text{because } S_b = V_b I_b$$

$$V = ZI \xrightarrow{\div V_b} V_{\text{p.u.}} = Z_{\text{p.u.}} I_{\text{p.u.}}$$

$$\frac{ZI}{V_b} = \frac{Z}{Z_b} \frac{I}{I_b} \quad \text{because } V_b = Z_b I_b$$

Normalization changes the scale while preserving the circuit relationships.

PART 3

Three-phase circuits

Two system bases, Y and delta branches, and preserved circuit laws

Two independent bases for Y and delta

Balanced three-phase network, with RMS voltage and current

Choose total apparent-power base

$$S_{3\phi,b}$$

$$\downarrow \div 3$$

$$S_{br,b} = S_{3\phi,b}/3$$

Power base for one Y or delta branch

Choose line-to-line voltage base

$$V_{LL,b}$$

$$\downarrow$$

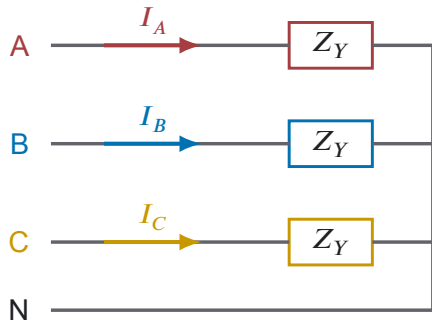
$$V_{Y,b} = V_{LL,b}/\sqrt{3}$$

$$V_{\Delta,b} = V_{LL,b}$$

The same two choices work for Y and delta. Derive all remaining bases, then divide each quantity by its matching base.

Y branch quantities and voltage references

One phase branch connects a line terminal to the star point N.



$$V_Y = V_{AN}, \quad I_Y = I_A$$

$$V_{AN} = Z_Y I_A$$

V_{AB} : line-to-line (phase-to-phase)

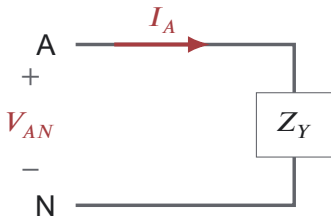
V_{AN} : line-to-neutral (Y phase)

$$|V_{AB}| = \sqrt{3} |V_{AN}| \text{ when balanced}$$

For a Y branch, use phase-to-neutral voltage and line current. These are the voltage and current of the same element.

Y branch: calculating the bases

Given $S_{3\phi,b}$ and $V_{LL,b}$



One branch, with power base $S_{3\phi,b}/3$.

$$S_{br,b} = \frac{S_{3\phi,b}}{3}, \quad V_{Y,b} = V_{LL,b}/\sqrt{3}$$

$$I_{Y,b} = \frac{S_{br,b}}{V_{Y,b}} = \frac{S_{3\phi,b}}{\sqrt{3}V_{LL,b}}$$

$$Z_{Y,b} = \frac{V_{Y,b}}{I_{Y,b}} = \frac{V_{LL,b}^2}{S_{3\phi,b}}$$

The four branch bases are ready. Normalize each voltage, current, impedance, or power by dividing by its matching base.

Y branch: normalized circuit laws

Ohm's law: $V_{AN} = Z_Y I_A$

$$(V_{Y,b} V_{AN,\text{p.u.}}) = (Z_{Y,b} Z_{Y,\text{p.u.}})(I_{Y,b} I_{A,\text{p.u.}})$$

Divide both sides by $V_{Y,b}$:

$$V_{AN,\text{p.u.}} = \frac{Z_{Y,b} I_{Y,b}}{V_{Y,b}} Z_{Y,\text{p.u.}} I_{A,\text{p.u.}}$$

Use $Z_{Y,b} I_{Y,b} = V_{Y,b}$:

$$V_{AN,\text{p.u.}} = Z_{Y,\text{p.u.}} I_{A,\text{p.u.}}$$

Branch power: $S_A = V_{AN} I_A^*$

$$(S_{\text{br},b} S_{A,\text{p.u.}}) = (V_{Y,b} V_{AN,\text{p.u.}})(I_{Y,b} I_{A,\text{p.u.}})^*$$

Divide both sides by $S_{\text{br},b}$:

$$S_{A,\text{p.u.}} = \frac{V_{Y,b} I_{Y,b}}{S_{\text{br},b}} V_{AN,\text{p.u.}} I_{A,\text{p.u.}}^*$$

Use $V_{Y,b} I_{Y,b} = S_{\text{br},b}$:

$$S_{A,\text{p.u.}} = V_{AN,\text{p.u.}} I_{A,\text{p.u.}}^*$$

Replace each physical quantity by its base times its per-unit value, then divide by the matching base.

Delta branch quantities and current directions

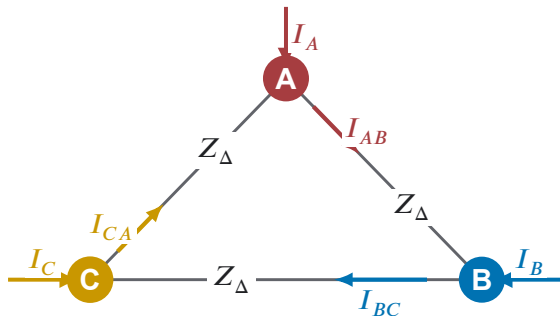
One physical branch: AB

$$V_{\Delta} = V_{AB}, \quad I_{\Delta} = I_{AB}$$

The branch sees line-to-line voltage and carries I_{AB} .

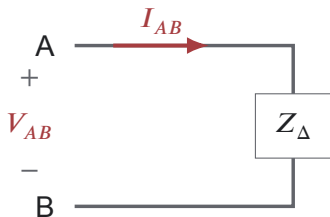
At terminal A, I_A and I_{CA} enter, while I_{AB} leaves.

$$I_A = I_{AB} - I_{CA}$$



Use the L08 directions: A to B, B to C, and C to A. Each delta branch sees a line-to-line voltage.

Delta branch: calculating the bases



Given $S_{3\phi,b}$ and $V_{LL,b}$

$$S_{br,b} = \frac{S_{3\phi,b}}{3}, \quad V_{\Delta,b} = V_{LL,b}$$

$$I_{\Delta,b} = \frac{S_{br,b}}{V_{\Delta,b}} = \frac{S_{3\phi,b}}{3V_{LL,b}}$$

$$Z_{\Delta,b} = \frac{V_{\Delta,b}}{I_{\Delta,b}} = \frac{3V_{LL,b}^2}{S_{3\phi,b}}$$

One branch, with power base $S_{3\phi,b}/3$.

The four branch bases are ready. Normalize each voltage, current, impedance, or power by dividing by its matching base.

Delta branch: normalized circuit laws

Ohm's law: $V_{AB} = Z_{\Delta} I_{AB}$

$(V_{\Delta,b} V_{AB,\text{p.u.}}) = (Z_{\Delta,b} Z_{\Delta,\text{p.u.}})(I_{\Delta,b} I_{AB,\text{p.u.}})$
Divide both sides by $V_{\Delta,b}$:

$$V_{AB,\text{p.u.}} = \frac{Z_{\Delta,b} I_{\Delta,b}}{V_{\Delta,b}} Z_{\Delta,\text{p.u.}} I_{AB,\text{p.u.}}$$

Use $Z_{\Delta,b} I_{\Delta,b} = V_{\Delta,b}$:

$$V_{AB,\text{p.u.}} = Z_{\Delta,\text{p.u.}} I_{AB,\text{p.u.}}$$

Branch power: $S_{AB} = V_{AB} I_{AB}^*$

$(S_{\text{br},b} S_{AB,\text{p.u.}}) = (V_{\Delta,b} V_{AB,\text{p.u.}})(I_{\Delta,b} I_{AB,\text{p.u.}})^*$
Divide both sides by $S_{\text{br},b}$:

$$S_{AB,\text{p.u.}} = \frac{V_{\Delta,b} I_{\Delta,b}}{S_{\text{br},b}} V_{AB,\text{p.u.}} I_{AB,\text{p.u.}}^*$$

Use $V_{\Delta,b} I_{\Delta,b} = S_{\text{br},b}$:

$$S_{AB,\text{p.u.}} = V_{AB,\text{p.u.}} I_{AB,\text{p.u.}}^*$$

Replace each physical quantity by its base times its per-unit value, then divide by the matching base.

Checking the delta–Y relationship

Physical impedances

$$Z_{\Delta} = 3Z_Y$$

Balanced delta and its equivalent Y.

Branch bases

$$Z_{\Delta,b} = 3Z_{Y,b}$$

Substitute base times per-unit impedance

$$Z_{\Delta,b}Z_{\Delta,\text{p.u.}} = 3Z_{Y,b}Z_{Y,\text{p.u.}}$$

$$(3Z_{Y,b})Z_{\Delta,\text{p.u.}} = 3Z_{Y,b}Z_{Y,\text{p.u.}}$$

Divide both sides by $3Z_{Y,b}$:

$$Z_{\Delta,\text{p.u.}} = Z_{Y,\text{p.u.}}$$

On recovery, $Z_{\Delta} = Z_{\Delta,b}Z_{\Delta,\text{p.u.}} = 3Z_{Y,b}Z_{Y,\text{p.u.}} = 3Z_Y$.

With these branch bases, the factor 3 is absorbed into the bases. The per-unit impedances are equal, and recovery preserves the physical relationship.

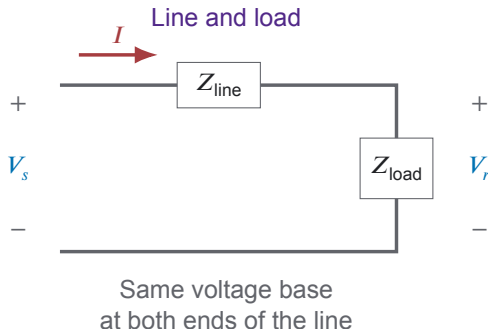
PART 4

Normalization and recovery

Specify bases, normalize, solve, and recover physical quantities

Normalization workflow

One voltage level with a stated phase or branch convention



1 Specify two independent bases

Power and voltage: S_b, V_b

2 Calculate the other bases

Current and impedance: I_b, Z_b

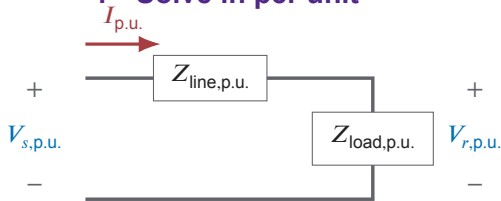
3 Normalize the known quantities

Convert the given circuit data to per unit

Choose two independent bases, derive the others, then normalize the given data.

Per-unit calculation and recovery

4 Solve in per unit



Use the normalized circuit laws

$$I_{p.u.} = \frac{V_{s,p.u.}}{Z_{line,p.u.} + Z_{load,p.u.}}$$

$$V_{r,p.u.} = Z_{load,p.u.} I_{p.u.}$$

5 Recover physical quantities

Multiply by the matching base

$$V = V_{p.u.} V_b$$

$$I = I_{p.u.} I_b$$

$$Z = Z_{p.u.} Z_b$$

$$S = S_{p.u.} S_b$$

One voltage level introduces the method. Its advantages become clearer with the multiple voltage levels in L14.

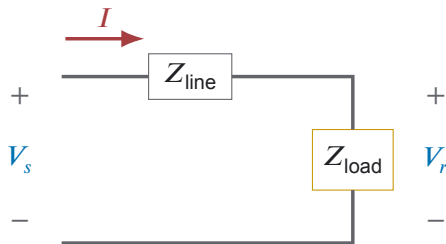
PART 5

Single-phase worked example

A source, line impedance, and load at one voltage level

Example: circuit data and specified bases

Single-phase circuit with RMS phasors



$$V_s = 240 \angle 0^\circ \text{ V}$$

$$Z_{\text{line}} = 0.36 + j0.72 \, \Omega$$

$$Z_{\text{load}} = 5.4 + j3.6 \, \Omega$$

1 Specify the independent bases

$$S_b = 10 \text{ kVA}$$

$$V_b = 240 \text{ V}$$

Find the line current, load voltage, and source, line, and load powers in per unit. Then recover their physical values.

Step 2: Derive the current base

$$I_b = \frac{S_b}{V_b} = \frac{10,000 \text{ VA}}{240 \text{ V}}$$

$$I_b = 41.6667 \text{ A}$$

The two specified bases determine the current base. It is a derived value, not another independent choice.

Step 2: Derive the impedance base

$$Z_b = \frac{V_b^2}{S_b} = \frac{(240 \text{ V})^2}{10,000 \text{ VA}}$$

$$Z_b = 5.7600 \, \Omega$$

$$S_b = 10 \text{ kVA}, \quad V_b = 240 \text{ V}, \quad I_b = 41.6667 \text{ A}, \quad Z_b = 5.76 \, \Omega$$

The complete base set is ready. Next, divide each circuit quantity by its matching base.

Step 3: Normalize the given data

Source voltage

$$V_{s,\text{p.u.}} = \frac{240\angle 0^\circ}{240}$$

$$V_{s,\text{p.u.}} = 1\angle 0^\circ$$

Line and load impedances

$$Z_{\text{line,p.u.}} = \frac{0.36 + j0.72}{5.76}$$

$$Z_{\text{line,p.u.}} = 0.0625 + j0.125$$

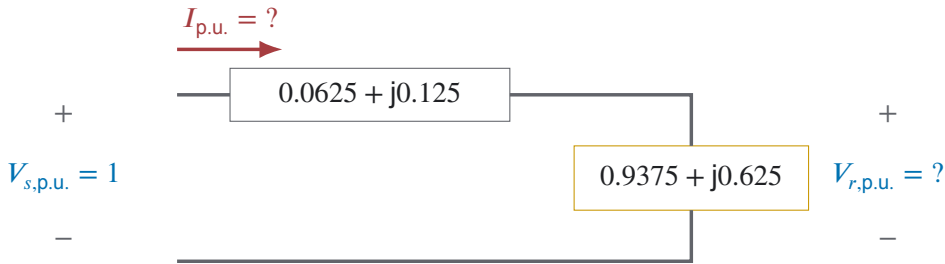
$$Z_{\text{load,p.u.}} = \frac{5.4 + j3.6}{5.76}$$

$$Z_{\text{load,p.u.}} = 0.9375 + j0.625$$

Normalize the known inputs first. Current, load voltage, and powers will be calculated from this per-unit circuit.

The circuit ready for per-unit calculation

The same circuit and reference directions, now in per unit



$$Z_{\text{total,p.u.}} = Z_{\text{line,p.u.}} + Z_{\text{load,p.u.}} = 1 + j0.75$$

The source drives the sum of the line and load impedances. Both impedances use the same base.

Step 4: Calculate the line current

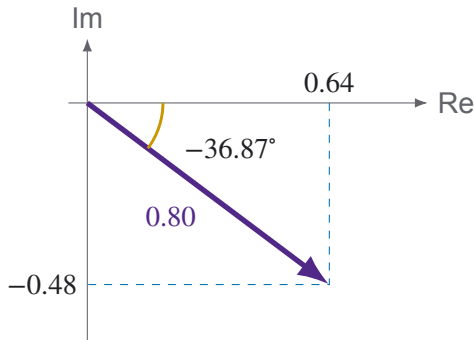
Solve the series circuit

$$I_{p.u.} = \frac{V_{s,p.u.}}{Z_{line,p.u.} + Z_{load,p.u.}}$$

$$= \frac{1}{1 + j0.75} = \frac{1 - j0.75}{1 + 0.75^2}$$

$$I_{p.u.} = 0.64 - j0.48$$

$$I_{p.u.} = 0.80 \angle (-36.87^\circ)$$



The computed current flows through both the line and the load. Its negative angle indicates lagging current.

Step 4: Calculate the load voltage

Voltage across the load

$$\begin{aligned} V_{r,\text{p.u.}} &= Z_{\text{load,p.u.}} I_{\text{p.u.}} \\ &= (0.9375 + j0.625)(0.64 - j0.48) \end{aligned}$$

$$\boxed{V_{r,\text{p.u.}} = 0.90 - j0.05}$$

Check with the line voltage drop

$$\begin{aligned} V_{\text{line,p.u.}} &= Z_{\text{line,p.u.}} I_{\text{p.u.}} \\ &= (0.0625 + j0.125)(0.64 - j0.48) \end{aligned}$$

$$= 0.10 + j0.05$$

$$V_{r,\text{p.u.}} = 1 - (0.10 + j0.05)$$

The load equation and KVL give the same voltage. The source and load share a base, while their operating voltages differ.

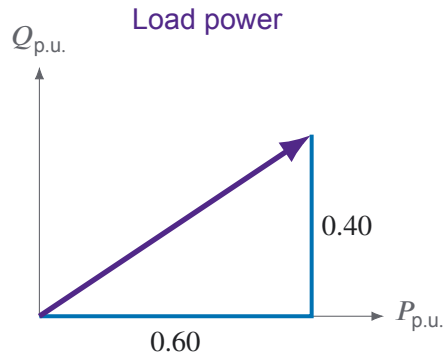
Step 4: Calculate the circuit powers

Use the solved voltages and current

$$\begin{aligned} S_{\text{load,p.u.}} &= V_{r,\text{p.u.}} I_{\text{p.u.}}^* \\ &= (0.90 - j0.05)(0.64 + j0.48) = 0.60 + j0.40 \end{aligned}$$

$$\begin{aligned} S_{\text{line,p.u.}} &= |I_{\text{p.u.}}|^2 Z_{\text{line,p.u.}} \\ &= 0.80^2(0.0625 + j0.125) = 0.04 + j0.08 \end{aligned}$$

$$S_{s,\text{p.u.}} = V_{s,\text{p.u.}} I_{\text{p.u.}}^* = 0.64 + j0.48$$



$S_{s,\text{p.u.}} = S_{\text{line,p.u.}} + S_{\text{load,p.u.}}$: **supplied complex power equals the power absorbed by the line and load.**

Step 5: Recover current and voltage

Line current in amperes

—

$$\begin{aligned} I &= I_{\text{p.u.}} I_b \\ &= (0.64 - j0.48)(41.6667 \text{ A}) \end{aligned}$$

$$I = 26.6667 - j20 \text{ A}$$

$$= 33.3333 \angle (-36.87^\circ) \text{ A}$$

Load voltage in volts

—

$$\begin{aligned} V_r &= V_{r,\text{p.u.}} V_b \\ &= (0.90 - j0.05)(240 \text{ V}) \end{aligned}$$

$$V_r = 216 - j12 \text{ V}$$

$$V_{\text{line}} = (0.10 + j0.05)(240) = 24 + j12 \text{ V}$$

Multiply each result by its matching base. The recovered voltage drops add to the 240-V source phasor.

Step 5: Recover the circuit powers

$$S = S_{\text{p.u.}} S_b, \quad S_b = 10 \text{ kVA}$$

Circuit quantity	Per-unit result	Recovered complex power
Source supplied power	$0.64 + j0.48$	$6.4 + j4.8 \text{ kVA}$
Line absorbed power	$0.04 + j0.08$	$0.4 + j0.8 \text{ kVA}$
Load absorbed power	$0.60 + j0.40$	$6.0 + j4.0 \text{ kVA}$

$$P_{\text{line}} = 0.4 \text{ kW}, \quad Q_{\text{line}} = 0.8 \text{ kvar}$$

The line consumes 400 W of real power. Real and reactive power both recover using the same apparent-power base.

Check: the recovered physical circuit

Direct calculation in physical units

$$Z_{\text{line}} + Z_{\text{load}} = 5.76 + j4.32 \, \Omega$$

$$I = \frac{240}{5.76 + j4.32} = 26.6667 - j20 \, \text{A}$$

$$\begin{aligned} V_r &= 240 - (0.36 + j0.72)I \\ &= 216 - j12 \, \text{V} \end{aligned}$$

Complex-power balance

$$S_s = V_s I^* = 6.4 + j4.8 \, \text{kVA}$$

$$\begin{aligned} S_{\text{line}} + S_{\text{load}} \\ &= (0.4 + j0.8) + (6.0 + j4.0) \\ &= 6.4 + j4.8 \, \text{kVA} \end{aligned}$$

The per-unit solution and the direct physical calculation describe the same source, line, and load.

Per-unit summary

Physical circuit

kV, A, Ω , MVA
one voltage level

divide by bases



Per-unit model

$$V_{\text{p.u.}} = I_{\text{p.u.}} Z_{\text{p.u.}}$$

$$S_{\text{p.u.}} = V_{\text{p.u.}} I_{\text{p.u.}}^*$$



multiply by bases

Choose S_{base} and V_{base} ; derive I_{base} and Z_{base} .

L13 develops the method at one voltage level. L14 shows its advantages across voltage levels linked by transformers.